

- 4 (a) (i) $F_E = Q_1 Q_2 / 4\pi\epsilon_0 r^2$ C1
 $= 8.99 \times 10^9 \times (1.6 \times 10^{-19})^2 / (2.0 \times 10^{-15})^2$
 $= 58 \text{ N}$ A1 [2]
- (ii) $F_G = Gm_1 m_2 / r^2$ C1
 $= 6.67 \times 10^{-11} \times (1.67 \times 10^{-27})^2 / (2.0 \times 10^{-15})^2$
 $= 4.7 \times 10^{-35} \text{ N}$ A1 [2]
- (b) (i) force of repulsion (much) greater than force of attraction B1
must be some other force of attraction M1
to hold nucleus together A1 [3]
- (Do not allow if $F_G > F_E$ in (a) or one of the forces not calculated in (a))*
- (ii) outside nucleus there is repulsion between protons B1
either attractive force must act only in nucleus
or if not short range, all nuclei would stick together B1 [2]